TIPA Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Efficient Multi-Agent Planning through Adaptive Reasoning Abstractions			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication			
	Medium Category	Small Category	Medium Category	Small Category
	Software	Embedded Software		
Concept of Proposed Technology	The proposed technology introduces an adaptive multi-agent planning and reasoning framework that enables efficient decision making by reusing and composing abstract reasoning units learned from prior executions. Instead of planning every task from scratch, the system decomposes execution logs into reusable abstractions, selectively caches high-utility reasoning patterns, and dynamically expands or refines them when fine-grained reasoning is required. The main functions of the technology are to extract and manage reusable reasoning abstractions, coordinate multiple agents operating at different levels of abstraction, and reduce computational cost and latency in complex planning and reasoning workflows. The technology can be used in multi-agent AI systems for data analytics, automated workflows, decision-support systems, and enterprise planning applications where repeated reasoning patterns occur and efficiency, scalability, and reliability are critical.			
Rationale for Joint R&D Project	• Complex planning and reasoning tasks in enterprise AI systems require repeated decomposition and coordination across multiple agents, leading to high computational cost and limited scalability when each task is planned from scratch. • Existing multi-agent planning solutions lack systematic mechanisms to reuse prior reasoning experience, resulting in redundant agent invocations, increased latency, and high deployment cost for real-world applications.			

	• There is a strong need for practical, deployable technology that improves the efficiency of multi-agent planning by introducing reusable abstractions while preserving reasoning accuracy and reliability. • The research institute contributes specialized expertise in multi-agent systems, abstraction learning, and efficient reasoning algorithms that are not typically available within small and medium-sized enterprises. • The Korean SME partner may provide real-world application scenarios, enterprise workflows, and operational constraints that are essential to shape the technology toward deployable and commercially relevant solutions. • Joint R&D enables rapid translation of research concepts into practical systems through iterative prototyping, pilot validation, and early alignment with product and market requirements. • Collaboration is necessary to reduce technical risk, accelerate development, and ensure that the resulting technology can transition efficiently from prototype to commercial deployment within the project timeframe.
Key Words	1. Multi-Agent Planning 2. Reasoning Abstractions 3. Efficient Decision Making 4. Adaptive Reasoning 5. Scalable AI Systems

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level 2	**End of the Project**	Level 4
R&D Objectives	• Develop a multi-agent planning and reasoning framework that learns and reuses abstract reasoning units to reduce redundant agent invocations and improve computational efficiency. • Design algorithms for extracting, pruning, and managing reusable reasoning abstractions from execution logs across heterogeneous planning tasks. • Build an adaptive orchestration layer that coordinates agents operating at different levels of abstraction and dynamically expands abstractions into fine-grained reasoning steps when required. • Implement abstraction selection and composition mechanisms that enable efficient reuse of prior reasoning patterns across related tasks and workflows. • Develop a deployable software prototype that integrates with existing multi-agent systems and enterprise workflows without requiring changes to underlying models or task definitions. • Validate the technology through industry-relevant planning and decision-making applications to ensure readiness for commercialization within the project timeframe.			

R&D Target Specifications	• Planning Efficiency and Cost Reduction
	  ○ Achieve 30–60% reduction in total agent invocations per task compared to baseline multi-agent planning without abstraction reuse.
	  ○ Reduce end-to-end planning and reasoning cost by 25–50% for representative multi-step workflows.
	• Runtime Performance
	  ○ Improve average task execution latency by 20–40% through reuse of cached abstract reasoning units.
	  ○ Limit orchestration and abstraction management overhead to ≤10% of total runtime per task.
	• Abstraction Quality and Reuse
	  ○ Achieve ≥50% reuse rate of cached abstract reasoning units across related tasks in benchmark workloads.
	  ○ Maintain task success rate within ≤2-3% degradation compared to full fine-grained planning from scratch.			

- Scalability
  - Support planning workflows involving 5–20 cooperating agents without significant degradation in throughput or latency.
  - Demonstrate stable performance for task sequences of 100+ consecutive planning queries under sustained load.
- System Robustness and Adaptivity
  - Automatically trigger fine-grained expansion for ≥90% of cases where abstract plans are insufficient or ambiguous.
  - Maintain execution failure rate below 5% across representative planning domains after online adaptation.
- Deployment Readiness
  - Deliver a production-ready software prototype that can be integrated into existing multi-agent platforms within 4-8 weeks.
  - Validate the technology through at least one industry-relevant pilot deployment by the end of the project.

R&D Plan

Overall Approach

The goal of this joint R&D project is to develop a practical multi-agent abstraction framework that improves the efficiency, scalability, and reliability of planning and reasoning in AI systems used for decision-making, automation, and data-centric workflows. The project follows a three-year, application-driven R&D approach, progressing from core abstraction learning to system integration and commercialization readiness.

Modern multi-agent planning systems rely heavily on repeated fine-grained reasoning and frequent large-model invocations, leading to high computational cost, increased latency, and limited scalability in real-world deployments [1, 2, 3]. Although hierarchical planning and macro-action methods have shown promise in controlled research settings, there is currently no deployable infrastructure that can automatically learn, manage, and reuse abstract reasoning units in production environments [4, 5]. This gap significantly limits the practical adoption of multi-agent planning systems in large-scale applications.

Motivating Example. Consider a multi-agent system answering the query: “Find the lowest-rated actor from the highest-rated movie in 1990.” A planner agent may decompose this into an abstract plan such as filter → sort → select → join → sort → select, which is then refined by worker agents into concrete operations. When a new but related query arrives, such as “Find the oldest actor in the highest-rated movie of 2003,” the same abstract plan can be reused by simply substituting parameters. This example illustrates how reusable abstract reasoning units can significantly reduce repeated planning, redundant agent invocations, and overall inference cost.

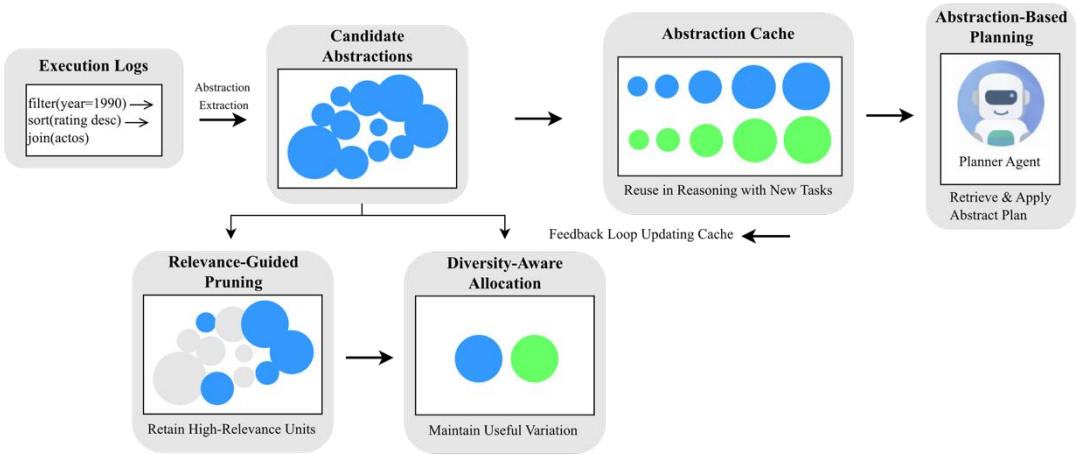

Figure 1. Overview of the pipeline for extracting and caching abstract reasoning units.

The proposed technology introduces a multi-agent abstraction layer in which agents operate at different levels of reasoning granularity. Frequently occurring reasoning patterns are collapsed into reusable abstract units, while fine-grained agents selectively expand these units only when necessary. This enables the system to reduce redundant planning, minimize expensive model calls, and adapt reasoning depth dynamically based on task complexity.

The project builds on the research team's prior work in compositional generalization [6], efficient reasoning [7], and long-context inference [8], and translates these ideas into an industry-ready planning and reasoning infrastructure. Figure 1 illustrates the high-level system architecture, showing how execution logs are decomposed into abstract reasoning units that are filtered, cached, and reused by a multi-agent planning system.

The R&D plan is organized into three phases aligned with increasing system maturity:
(1) abstraction learning and relevance-guided pruning,
(2) diversity-aware allocation and reasoning with abstractions, and
(3) system integration and commercialization readiness.

Year 1: Abstraction Learning and Relevance-Guided Pruning

The first year focuses on establishing the foundational abstraction learning framework and delivering an initial prototype capable of extracting and pruning abstract reasoning units from multi-agent execution logs. During this phase, the project team will design methods to decompose execution traces into candidate reasoning units at multiple levels of granularity, ranging from atomic actions to coarse macro-actions. For example, in the running example above, atomic steps such as filter by year or sort by rating are grouped into higher-level abstract plans that can be reused across similar queries.

A core objective of Year 1 is to develop relevance-guided pruning mechanisms that identify which abstractions are worth caching. Not all extracted reasoning units provide long-term value; some are rare, overly specific, or redundant. The system will assign relevance scores based on frequency, contribution to task performance, and expected reuse, and will remove low-utility units to prevent cache growth and performance degradation.

Methodologically, this phase will combine offline analysis of historical execution logs with online instrumentation of multi-agent systems to collect representative planning trajectories. Relevance scoring functions will be implemented and evaluated on benchmark planning and reasoning tasks.

Key milestones for Year 1 include:

- A prototype abstraction extraction and pruning pipeline.
- Initial relevance metrics and pruning policies validated on representative tasks.
- A cached abstraction repository integrated into a basic multi-agent planner.

Figure 2 illustrates the abstraction extraction and pruning pipeline, showing how execution logs are decomposed into candidate abstractions and filtered before caching. The primary outcome of Year 1 is a working prototype that demonstrates measurable reductions in redundant agent invocations on repeated or related tasks.

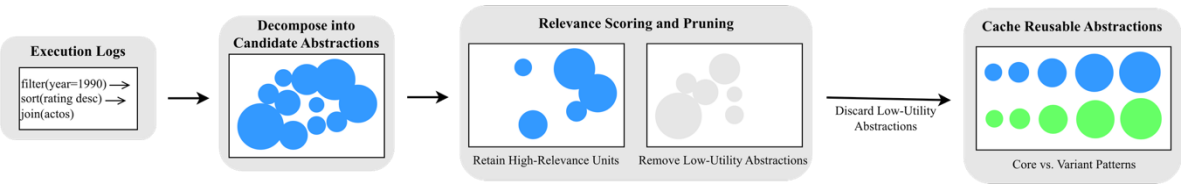

Figure 2. Abstraction extraction and pruning pipeline where execution logs are decomposed into candidate abstractions, scored for relevance, pruned to retain high-utility units, and stored in abstraction cache for reuse in future tasks.

Year 2: Diversity-Aware Allocation and Reasoning with Abstractions

The second year centers on improving generalization and robustness by regulating how abstractions are allocated across abstraction levels and reasoning trajectories, and by developing execution pipelines that map abstract units back to concrete actions.

Building on the running example, while a single abstract plan such as filter → sort → select → join → sort → select can be reused across many queries, variations in data distributions and task goals require preserving diversity among abstract plans. The project will develop diversity-aware allocation mechanisms that maintain multiple variants of high-level abstractions while limiting redundancy at lower levels.

During this phase, the team will design both online greedy strategies and offline learned models to decide which abstractions to retain, merge, or discard as new execution data becomes available. In parallel, execution pipelines will be developed to compose cached abstractions into executable plans for new tasks, with adaptive expansion when finer-grained reasoning is required to preserve correctness.

Online monitoring and feedback mechanisms will be introduced to detect execution failures and trigger refinement of cached abstractions. For example, if a reused abstract plan produces systematic errors on a new class of queries, the system will refine or split the abstraction accordingly.

Key milestones for Year 2 include:

- A diversity-aware abstraction allocation module integrated into the cache management system.
- An abstraction-based execution pipeline supporting adaptive expansion and online feedback.
- Validation of reduced agent invocations and stable performance across diverse reasoning benchmarks.

Figure 3 summarizes the end-to-end reasoning workflow, from abstract plan retrieval to adaptive execution and feedback-driven refinement. The primary outcome of Year 2 is a robust abstraction-based planning system that demonstrates improved efficiency while preserving accuracy and adaptability.

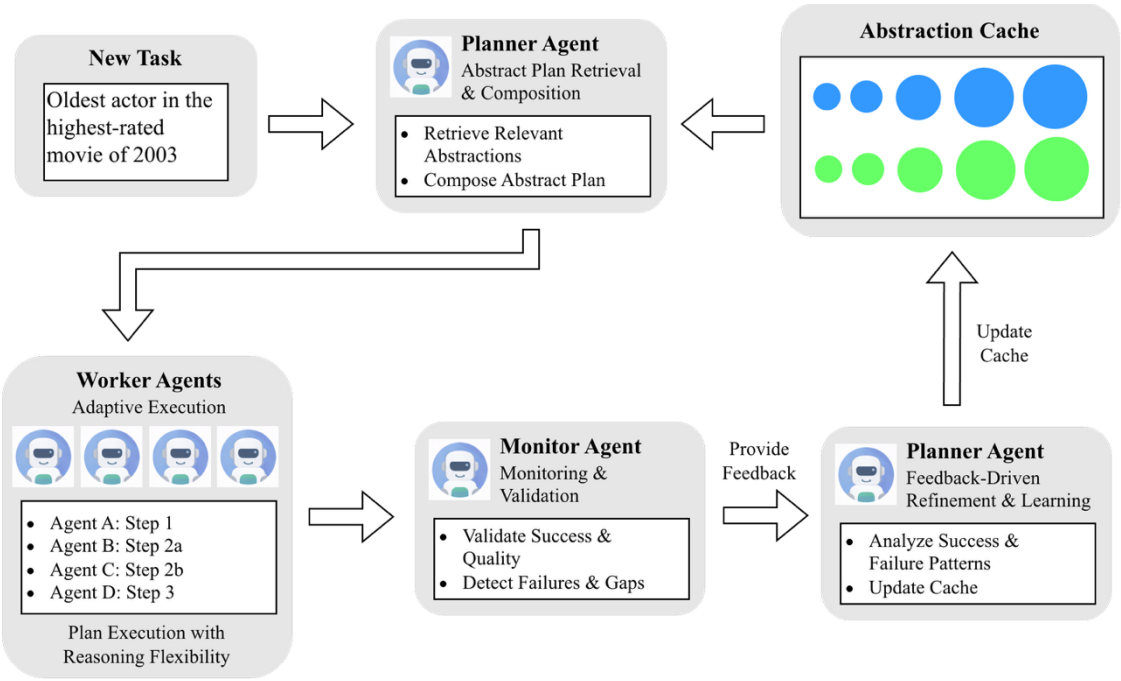

Figure 3. End-to end reasoning workflow: From a new task, the Planner Agent retrieves abstract plans from the cache, Worker Agents execute them adaptively while being validated by the Monitor Agent, and the Updater Agent refines the cache based on feedback signals.

	Year 3: System Integration and Commercialization Readiness The third year focuses on preparing the abstraction framework for deployment in industrial multi-agent systems and for commercialization. During this phase, the team will integrate the abstraction layer into industry-relevant planning and reasoning platforms, such as data-centric workflows, decision-support systems, and enterprise automation pipelines. Emphasis will be placed on engineering maturity, scalability, usability, and reliability under real-world workloads. User-facing APIs, configuration tools, and monitoring dashboards will be developed to allow practitioners to control abstraction granularity, caching policies, and execution strategies. Extensive system-level testing will evaluate performance gains, latency impact, robustness, and maintainability. In collaboration with industry partners, the team will identify commercialization pathways, including licensing, product integration, and potential startup or spin-off opportunities. Key milestones for Year 3 include: • A production-ready abstraction management and execution framework. • Successful pilot deployments in at least one industry application. • Complete documentation, deployment guides, and commercialization plans. The primary outcome of Year 3 is a deployable multi-agent abstraction infrastructure that significantly reduces planning and reasoning cost while preserving correctness and flexibility. **Expected Milestones and Timeline Summary** Across the 3 years, the project will progress through: • Year 1: Abstraction extraction, relevance-guided pruning, prototype cache system. • Year 2: Diversity-aware allocation, abstraction-based execution, adaptive refinement. • Year 3: System integration, pilot validation, commercialization readiness. This phased approach ensures steady risk reduction, early demonstration of efficiency gains, and a clear path from research prototype to industrial deployment.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the technical lead and primary developer of the proposed multi-agent abstraction technology. Its role is to provide the core scientific, algorithmic, and system-level expertise required to design, implement, and validate an efficient planning and reasoning framework suitable for industrial deployment. **Technical Leadership and Core Algorithm Development** The research institute will lead the overall technical design and system architecture of the abstraction-based planning framework. This includes developing algorithms for extracting reusable reasoning abstractions from execution logs, designing relevance-guided pruning and diversity-aware allocation mechanisms, and defining orchestration strategies for coordinating agents across multiple abstraction levels. **Research Scope and Methodology** The institute will conduct the applied research necessary to translate abstraction and multi-agent reasoning concepts into a deployable system. This includes studying trade-offs between abstraction granularity and reasoning fidelity, developing adaptive refinement mechanisms, and establishing evaluation methodologies to measure efficiency, reuse rate, and task performance.

	Prototype Implementation and System Integration The institute will implement and maintain the core software components of the proposed technology, including the abstraction management layer, agent orchestration module, and execution monitoring framework. It will provide reference implementations, APIs, and integration guidelines to support deployment within the SME partner's multi-agent systems and enterprise workflows. **Performance Evaluation and Validation** The institute will lead the design and execution of experimental evaluations, including benchmarking planning efficiency, latency, cost reduction, abstraction reuse, and success rate across representative planning domains. These evaluations will guide iterative system refinement and document measurable performance gains required for commercialization readiness. **Resource Provision and Infrastructure Support** The research institute will provide the necessary computational resources, software infrastructure, and research personnel to carry out the project. This includes access to development servers, simulation environments, and testing frameworks for multi-agent planning experiments. Faculty researchers, graduate students, and dedicated research engineers will contribute to system development, experimentation, and documentation. **Knowledge Transfer and Commercialization Support** The institute will play a central role in knowledge transfer and technology maturation by preparing technical documentation, developer guides, and training materials to facilitate adoption by industry partners. It will collaborate with the SME partner on intellectual property management, licensing strategies, and potential technology transfer or startup formation.
Expected R&R of Korean SME Partner	The Korean SME partner will serve as the primary industrial collaborator and commercialization driver for the proposed joint R&D project. Its role is to provide real-world application contexts, development resources, and product integration pathways that ensure the proposed multi-agent abstraction technology addresses practical industry needs and can transition efficiently into commercial products. **Application Definition and Use-Case Leadership** The SME partner will identify and define industry-relevant planning and reasoning use cases where multi-agent coordination and efficiency are critical, such as enterprise workflow automation, data analytics pipelines, decision-support systems, and intelligent orchestration platforms. These use cases will guide system requirements, performance targets, and deployment constraints throughout the project. **System Integration and Software Development** The SME partner will lead the integration of the proposed technology into its existing multi-agent platforms and product pipelines. This includes adapting application interfaces, configuring deployment environments, embedding the abstraction management and orchestration modules into production workflows, and developing application-level components required for end-to-end operation. **Testing, Validation, and Pilot Deployment** The SME partner will play a central role in testing and validating the technology under real-world operating conditions. This includes conducting pilot deployments, executing functional and performance testing, collecting operational metrics, and evaluating efficiency gains, reliability, and

	user experience. Feedback from these activities will guide iterative refinement toward production readiness. **Resource Commitment and Infrastructure Support** The SME partner will provide dedicated engineering personnel, application environments, and deployment infrastructure required for development and validation. This includes access to representative datasets, enterprise workflows, cloud or on-premise systems, and monitoring tools used in operational settings. Product managers and system engineers will participate actively in joint development and evaluation activities. **Commercialization and Market Deployment** The SME partner will take a leading role in commercialization planning and market deployment. This includes defining product offerings that incorporate the developed technology, identifying target customer segments, preparing go-to-market strategies, and planning product release schedules. The partner will collaborate with the research institute to explore licensing, product integration, and potential commercialization pathways. **Collaboration and Project Coordination** The SME partner will participate actively in joint project planning, regular technical meetings, milestone reviews, and progress reporting. It will coordinate closely with the research institute to align technical development with product timelines and business objectives, ensuring efficient progress toward commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** The primary technological outcome of this joint R&D project will be a deployable multi-agent planning and reasoning infrastructure that significantly improves the efficiency and scalability of complex decision-making systems. The project will deliver a production-ready software framework capable of learning, managing, and reusing abstract reasoning units to reduce redundant planning, lower computational cost, and maintain high task performance. Key technical deliverables include a modular abstraction management layer, an adaptive agent orchestration module, and validated benchmarks demonstrating measurable improvements in planning latency, agent invocation reduction, and abstraction reuse. These outcomes establish a practical foundation for efficient multi-agent reasoning in real-world enterprise environments. **Commercialization Outcomes** A central outcome of the project is the transition of the developed technology from prototype to commercial deployment. Through close collaboration with the Korean SME partner, the project will deliver validated pilot systems and a production-ready planning framework suitable for integration into commercial multi-agent platforms and enterprise software products. Expected commercialization results include incorporation of the technology into the SME partner's product portfolio, development of market-ready AI solutions with improved efficiency and lower operational cost, and establishment of licensing or product integration pathways. These outcomes create direct opportunities for product differentiation, new revenue streams, and expansion into customer segments that require scalable and cost-efficient intelligent systems.

The project also establishes a foundation for broader commercialization channels, including technology licensing to additional SMEs, adoption by AI platform providers, and potential startup or spin-off activities centered on efficient multi-agent planning infrastructure.

Industrial Impact

The project addresses a critical bottleneck in industrial AI systems: the high cost and limited scalability of complex planning and reasoning workflows in multi-agent environments. By enabling systematic reuse of prior reasoning experience, the proposed technology supports broader deployment of intelligent automation across sectors such as enterprise software, data analytics, workflow management, logistics, and decision-support systems.

Expected industrial impacts include:

- Enabling more efficient deployment of multi-agent systems in large-scale enterprise applications.
- Improving competitiveness of SMEs by allowing them to offer more capable intelligent products at lower cost.
- Reducing infrastructure requirements and operational overhead for complex planning and orchestration tasks.

These impacts contribute to a more scalable and accessible ecosystem for advanced AI-driven decision systems.

Economic and Ecosystem Impact

The project contributes to economic growth and innovation by strengthening the planning and reasoning infrastructure capabilities of Korean SMEs and supporting the creation of high-value AI products and services. By lowering deployment cost and improving system efficiency, the technology reduces entry barriers for AI-driven businesses and supports job creation in AI engineering, software development, and intelligent automation.

At the ecosystem level, the project fosters a sustainable academia–industry collaboration model that accelerates technology transfer from research to market. The outcomes strengthen national competitiveness in intelligent systems and multi-agent AI technologies and support the development of a robust AI innovation ecosystem with global relevance.

Long-Term Impact

Beyond the project horizon, the developed abstraction-based planning framework establishes a long-term platform for efficient reasoning applicable to future generations of autonomous agents and intelligent systems. The infrastructure can be extended to new application domains, larger agent populations, and more complex decision environments, ensuring lasting relevance as AI systems continue to grow in scale and sophistication.

Overall, the joint R&D project delivers both a commercially viable planning technology and a strategic contribution to scalable AI adoption, enabling sustained industrial innovation and economic impact.

6. References

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□ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization